

1 · RECOMMENDATION

*Meet Mrs. D, 71. Discharged home on a Friday with a new insulin dose.
Eighteen days later she is back through the ED — her third admission this year.*

Illustrative composite patient. Her profile is the segment on the next slide.

**Don't build AI yet. Flag the frequent
returners with one rule, and fix
their discharge.**

22.0%

30-day readmission, 2+ prior stays

11.4%

everyone else's baseline

Diabetes 130-US Hospitals, 1999–2008 · 99,343 eligible encounters (death and hospice discharges excluded)

~90 seconds. Open with Mrs. D, pause, then land the recommendation. The ask: approve a 90-day rules-plus-process pilot, not an ML build. Mrs. D is an illustrative composite, not a real record; say so if asked. 22.0% is the 30-day rate for encounters with 2+ inpatient stays in the prior year; 11.4% is the eligible-encounter baseline.

2 · EVIDENCE

Mrs. D isn't rare: risk follows prior admissions, not diagnosis

Prior-year stays	Encounters	30-day rate	Share of readmits
0	66.7%	8.6%	50.3%
1	19.1%	13.3%	22.2%
2	7.3%	17.9%	11.6%
3 or more	6.9%	26.4%	15.9%

Segment (2+ stays, shaded): 14.2% of encounters, **27.5% of readmissions**. By primary diagnosis, every group sits between 9.5% and 13%.

27.5%

when the primary diagnosis is diabetes itself

34.7%

discharged to rehab (small group, n=262)

12% vs 17%

A1C tested in the segment vs overall: a process gap

Unit: inpatient encounter · outcome: readmitted <30 days · 99,343 eligible encounters, 11,314 readmissions · patient-level first-encounter rate 9.0%

~2 minutes. One row = one encounter; outcome = readmitted within 30 days, because that is what payers penalize and what discharge care can affect. Excluded discharges to death or hospice (IDs 11, 13, 14, 19, 20, 21). If asked "patients or encounters?": encounter rate 11.4%; first encounter per patient 9.0%. Segment = 14,114 encounters, 7,901 patients, 3,108 readmissions. Patterns inside the segment: diabetes primary dx 27.5% (n=1,674), rehab 34.7% (n=262, small), A1C tested 12.3% vs 16.9% overall.

3 · BUSINESS VALUE AND BREAK-EVEN

Targeting halves the bar the intervention must clear

Required relative reduction =
**cost per patient ÷ (readmission rate × value of one
avoided readmission)**

Per 1,000 discharges	Segment only	Everyone
Expected readmits	220	114
Program cost	\$400k	\$400k
Avoided to break even	33	33
Relative cut needed	~15%	~29%

Range if one avoided readmission is worth \$8k–\$15k: the segment needs a 12%–23% cut.

Who captures it?

Under fee-for-service, an avoided readmission is lost revenue for the hospital.

Value reaches the hospital only through penalties avoided, value-based or capitated contracts, or freed bed capacity.

The data holds no cost, revenue or payment rules: CFO to confirm.

ASSUMPTIONS, not in data: \$400 per patient (follow-up + med reconciliation), \$12,000 per avoided readmission · rates are FACTS from the dataset

~1.5 minutes. Walk the formula once. Segment: $400 \div (0.22 \times 12,000) = 15.2\%$. Everyone: $400 \div (0.114 \times 12,000) = 29.2\%$. $\$400k \div \$12k = 33.3$ avoided readmissions to break even. Sensitivity: $400 / (0.22 \times 15,000) = 12.1\%$; $400 / (0.22 \times 8,000) = 22.7\%$. CFO pushback to expect: "We'd lose revenue." Answer: yes under fee-for-service; that is exactly why contract mix is unknown #1 on slide 5.

4 · INTERVENTION

A one-line rule gets most of what a model would

Process ✓

Rules ✓

Analytics

ML

GenAI

Agentic AI

Flag top 14%	Rule: 2+ stays	Boosted model
Precision	21.7%	25.0%
Readmits caught	26.6%	30.7%
AUC	0.60	0.67

Model gain = 33 more readmits caught per 1,000 flagged, on 2008 data it would need retraining for.

What Mrs. D gets

- Flagged at discharge: 2+ stays in a year
- Follow-up visit within 7 days
- Medication reconciliation call
- AIC checked before she leaves

Test set split by patient (25%, seed 42) · model: gradient boosting on all fields except weight · rule scored by prior inpatient count

~1.5 minutes. Ladder: we stop at Process + Rules. Same 14% flag rate for both. Rule precision 21.7%, recall 26.6%, AUC 0.602; GBM precision 25.0%, recall 30.7%, AUC 0.674. The gap is about 33 extra readmissions caught per 1,000 flagged, and the model would need retraining on current local data plus monitoring. The rule runs today from the admission record. ML becomes worth revisiting when the pilot proves the intervention works and rule precision becomes the bottleneck.

5 · UNKNOWNNS AND VALIDATION PLAN

Validate two things in 90 days before anything gets built

Type	Claim	How we close it
Fact	2+ prior stays: 22.0% rate vs 11.4%	Reproducible script
Assumption	\$12k value per avoided readmit, \$400 cost	CFO: current contract mix
Assumption	2008 pattern holds in our hospitals	Rerun rule on 12 months local data
Hypothesis	7-day follow-up cuts segment readmits $\geq 15\%$	90-day pilot vs matched control

Go / no-go at day 90

Scale if the cut clears break-even under our real contracts.
Revisit ML only if rule precision becomes the bottleneck.

*For Mrs. D, the fix is a clinic slot and a phone call,
not a model.*

Data gaps: no cost or hospital ID · weight 97% missing, A1C 83%, specialty 49% · association, not proof of preventability

~1.5 minutes, then into 4 minutes of pushback. Ask: approve the 90-day pilot and the two validations. Close the loop on Mrs. D. Pushback prep: no hospital identifier means no per-site sizing; data predates the readmission penalty era; prior admissions predicting readmission does not prove the follow-up prevents it, which is why the pilot needs a matched control.